

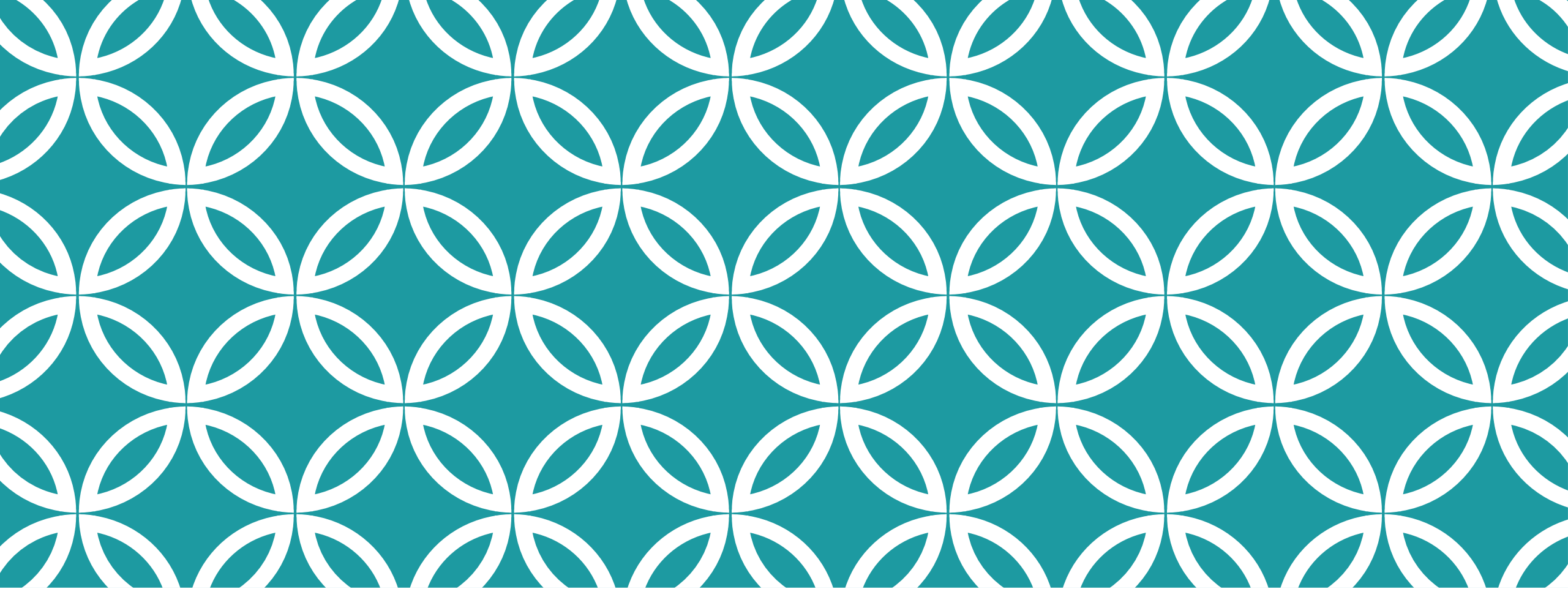


FLAME
UNIVERSITY

PUBP 233 INTRODUCTION TO DEMOGRAPHY (POPULATION AND SOCIETY)

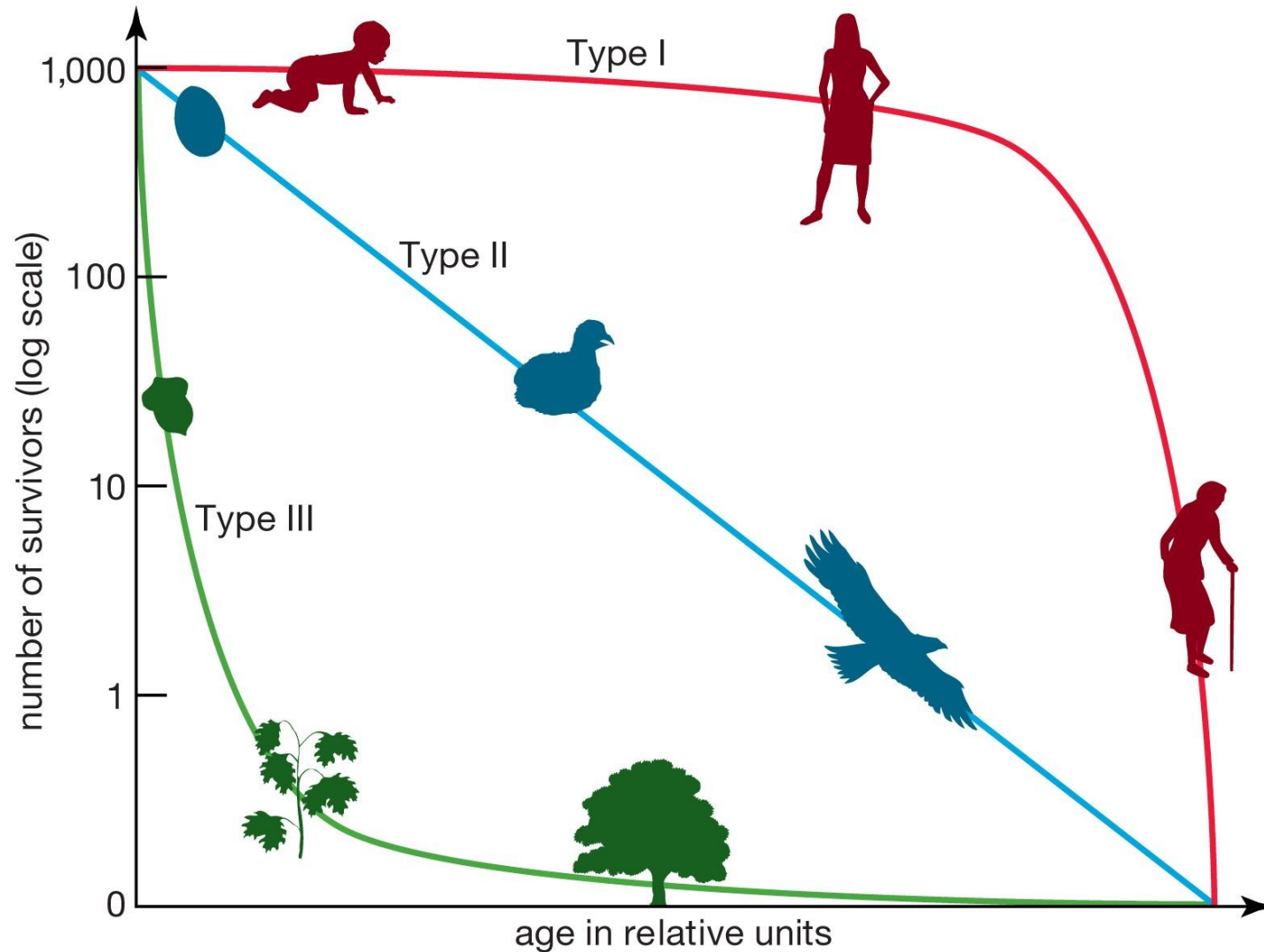
Week 6: Life Table

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LIFE TABLE |

GENERIC CONCEPT



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<https://www.britannica.com/science/population-ecology/Life-tables-and-the-rate-of-population-growth>

LIFE TABLE

Experiences of an artificial cohort experiencing the vital rates of the population

Summarizes the mortality experience of a population

Age specific death rates constant over time, flow of births is constant over time

Stationary population , Net immigration is zero

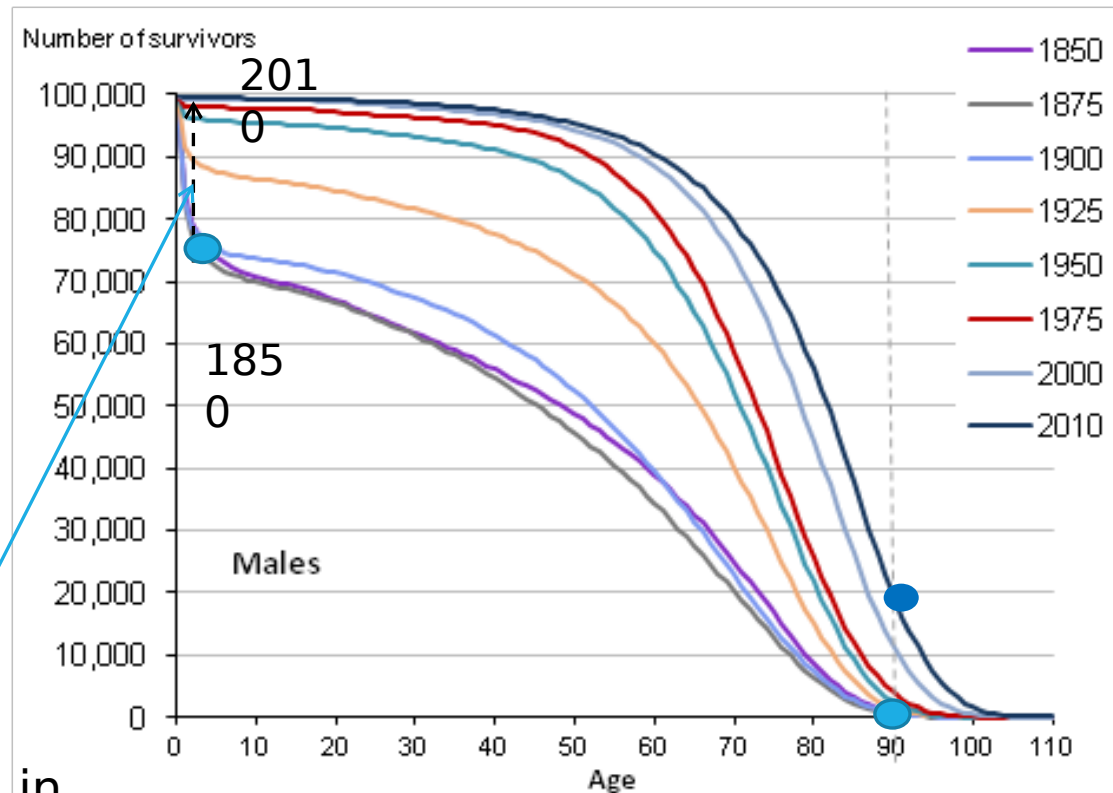
Introduction to Demographic Methods : Lecture Materials, Johns Hopkins University,
<http://ocw.jhsph.edu/courses/demographicmethods/lectureNotes.cfm>

COLUMNS OF A LIFE TABLE

Country		Year							
India		2005-2010							
Age(x)	$m(x,n)$	$a(x,n)$	$q(x,n)$	$p(x,n)$	$l(x)$	$d(x,n)$	$L(x,n)$	$T(x)$	$e(x)$
0	0.05338	0.20	0.05120	0.94880	100 000	5 120	95 922	6 668 988	66.69
1	0.00424	1.44	0.01677	0.98323	94 880	1 591	375 449	6 573 065	69.28
5	0.00137	2.50	0.00684	0.99316	93 289	638	464 851	6 197 616	66.43
10	0.00102	2.50	0.00508	0.99492	92 651	470	462 080	5 732 765	61.87
15	0.00168	2.65	0.00839	0.99161	92 181	773	459 091	5 270 685	57.18
20	0.00218	2.55	0.01083	0.98917	91 408	990	454 609	4 811 594	52.64
...	...	...	...	...	...	...	...	...	...
85	0.17852	5.60	1.00000	0.00000	17 448	17 448	97 736	97 736	5.60

SURVIVORSHIP

l_x = probability of surviving till age x



$$p(x) = \frac{l_x}{l_0}$$

$$l_0 = 100,000$$

Reduction in
infant
mortality

Mortality in England and Wales: Average Life Span

<http://www.ons.gov.uk>

OTHER USES OF LIFE TABLE

Survival in marriage (before divorce or death)

time to re-offending on release from prison

Birth spacing

time to first sex ("virginity table")

Survival in school

cancer survival

Survival of bulbs/laptop/manufactured good before failure

A criminologist could use a life table to make sense of data on lengths of time after a person is released from prison before re-offending occurs. In this case the start event is release from prison and the end event would be a first re-offence.

The l_x column would represent the numbers free from re-offending, from a radix of, say 1000, by numbers of months or years since release. The concept is exactly the same as for mortality and the life table would be constructed in the same way. In this example the duration would be measured in units of time, not of age.

MODEL COHORT LIFE TABLE OF 10 PEOPLE

Table 3.1: Life table for hypothetical cohort of 10 births shown in figure 3.1

Exact age x	Number left alive at age x l_x	Number dying between ages x and $x + n$ ${}_n d_x$	Probability of dying between ages x and $x + n$ ${}_n q_x$	Probability of surviving from age x to age $x + n$ ${}_n p_x$	Person-years lived between ages x and $x + n$ ${}_n L_x$	Person-years lived above age x $T_x = \sum_{a=x}^{\infty} {}_n L_a$	Expectation of life at age x $e_x^0 = T_x / l_x$	Death rate in the cohort between ages x and $x + n$ ${}_n m_x$	Average person-years lived in the interval by those dying in the interval ${}_n a_x$
0	10	1	1/10	9/10	$9 + .07 = 9.07$	$436.79 + 9.07 = 445.86$	$\frac{445.86}{10} = 44.586$	$\frac{1}{9.07}$	.07
1	9	1	1/9	8/9	$8 \cdot 4 + .22 = 32.22$	$404.57 + 32.22 = 436.79$	$\frac{436.79}{9} = 48.532$	$\frac{1}{32.22}$	.22
5	8	0	0	1	$8 \cdot 5 = 40$	$364.57 + 40 = 404.57$	$\frac{404.57}{8} = 50.571$	0	—
10	8	1	1/8	7/8	$7 \cdot 10 + 6.41 = 76.41$	$288.16 + 76.41 = 364.57$	$\frac{364.57}{8} = 45.571$	$\frac{1}{76.41}$	6.41
20	7	1	1/7	6/7	$6 \cdot 10 + 2.12 = 62.12$	$226.04 + 62.12 = 288.16$	$\frac{288.16}{7} = 41.166$	$\frac{1}{62.12}$	2.12

RATE AND PROBABILITY

$$\text{Rate} = \frac{\text{Number of Occurrences}}{\text{Number of Person-years Lived}}$$

$$\text{Probability} = \frac{\text{Number of Occurrences}}{\text{Number of Preceding Events or Trials}}$$

The concept of “probability” is used as a step in life table construction to get the expected number of survivors in a cohort

Probability that a marriage would end in a divorce for a given birth cohort by counting, over all **members of the cohort**, the number of marriages and the number of divorces over the cohort’s lifetime:

$$q^D = \frac{\text{Number of Divorces}}{\text{Number of Marriages}}$$

$$0 \leq q^D \leq 1$$

“relative frequency” approach to estimating the probability of divorce.

$$0 \leq p \leq 1$$

x	n	${}_nq_x$	${}_np_x$	l_x	${}_nd_x$	${}_nL_x$	T_x	e_x
0	1	0.008252	0.991748	100 000	825	99 258	7 756 261	77.563
1	4	0.001630	0.998370	99 175	162	396 311	7 657 003	77.207
5	5	0.000905	0.999095	99 013	89	494 842	7 260 692	73.331
10	5	0.000935	0.999065	98 924	93	494 388	6 765 850	68.394
15	5	0.001409	0.998591	98 831	139	493 808	6 271 462	63.456
20	5	0.001534	0.998466	98 692	152	493 080	5 777 654	58.542
25	5	0.001818	0.998182	98 540	179	492 253	5 284 574	53.629
30	5	0.002826	0.997174	98 361	278	491 110	4 792 321	48.722
35	5	0.004410	0.995590	98 083	432	489 335	4 301 211	43.853
40	5	0.007199	0.992801	97 651	693	486 523	3 811 876	39.036
45	5	0.012348	0.987652	96 958	1 197	481 798	3 325 353	34.297
50	5	0.020831	0.979169	95 761	2 005	473 793	2 843 555	29.694
55	5	0.035455	0.964545	93 756	3 324	460 470	2 369 762	25.276
60	5	0.058507	0.941493	90 432	5 291	438 933	1 909 292	21.113
65	5	0.087310	0.912690	85 141	7 434	407 120	1 470 359	17.270
70	5	0.139189	0.860811	77 707	10 816	361 495	1 063 239	13.683
75	5	0.220993	0.779007	66 891	14 782	297 500	701 744	10.492
80	5	0.352367	0.647633	52 109	18 362	214 640	404 244	7.758
85		1.000000	0.000000	33 747	33 747	189 604	189 604	5.618

Table 1: A typical mortality abridged life table .

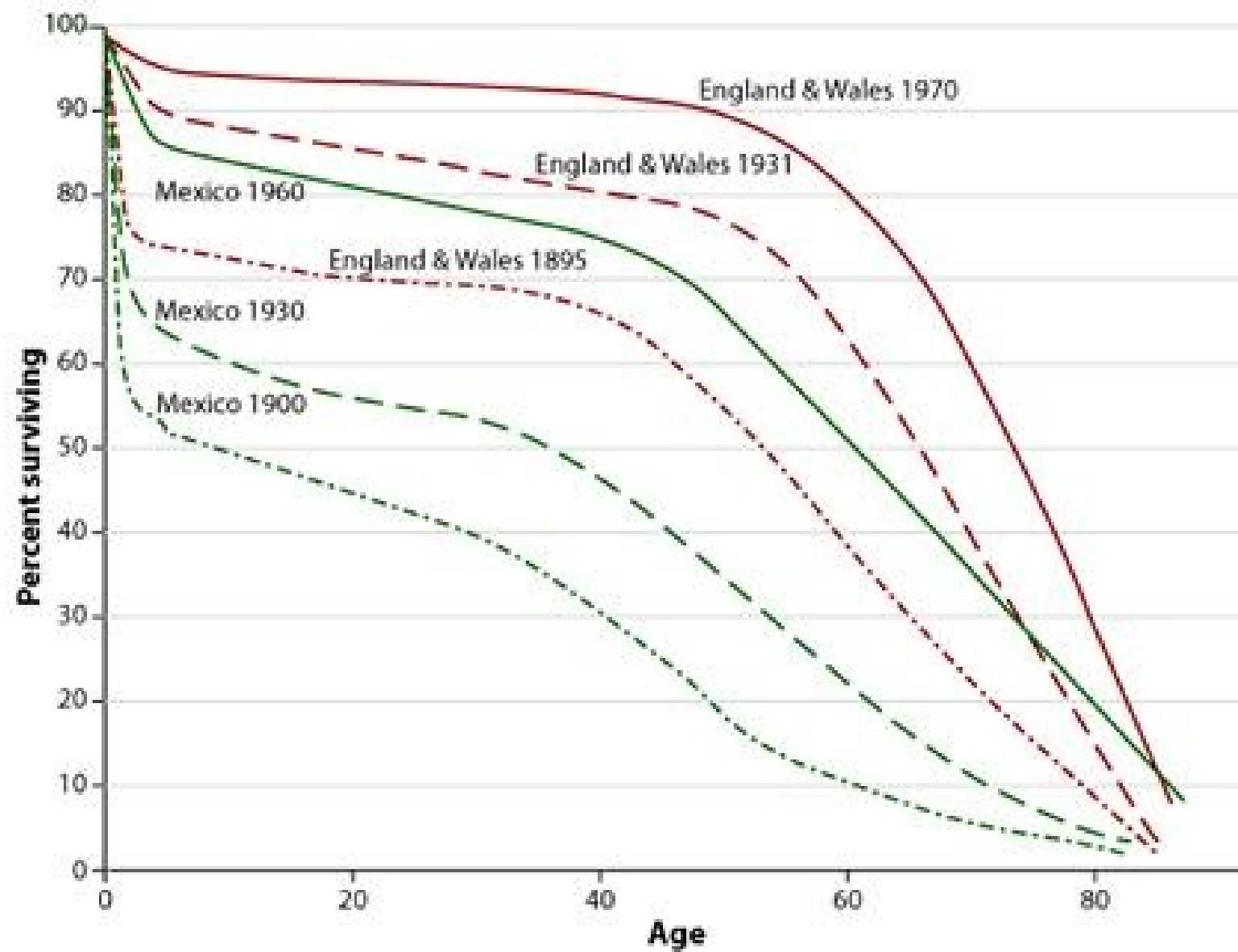
Source: Office for National Statistics

EDMOND HALLEY-LIFE TABLE

Age. Curt.	Per- sons.	Age. Curt.	Per- sons.	Age. Curt.	Per- sons.	Age. Curt.	Per- sons.	Age. Curt.	Per- sons.	Age. Curt.	Per- sons.	Age.	Persons.
1	1000	8	680	15	628	22	585	29	539	36	481	7	5547
2	855	9	670	16	622	23	579	30	531	37	472	14	4584
3	798	10	661	17	616	24	573	31	523	38	463	21	4270
4	760	11	653	18	610	25	567	32	515	39	454	28	3964
5	732	12	646	19	604	26	560	33	507	40	445	35	3604
6	710	13	640	20	598	27	553	34	499	41	436	42	3178
7	692	14	634	21	592	28	546	35	490	42	427	49	2709
Age. Curt.	Per- sons.	Age. Curt.	Per- sons.	Age. Curt.	Per- sons.	Age. Curt.	Per- sons.	Age. Curt.	Per- sons.	Age. Curt.	Per- sons.	Age.	Persons.
43	417	50	346	57	272	64	202	71	131	78	58	56	2194
44	407	51	335	58	262	65	192	72	120	79	49	63	1694
45	397	52	324	59	252	66	182	73	109	80	41	70	1204
46	387	53	313	60	242	67	172	74	98	81	34	77	692
47	377	54	302	61	232	68	162	75	88	82	28	84	253
48	367	55	292	62	222	69	152	76	78	83	23	100	107
49	357	56	282	63	212	70	142	77	68	84	20		
												34000	
												Sum Total.	

Edmond Halley's Life Table of 1693. Note that the Radix is 1000.

"He laid the foundation for actuarial



AGE SPECIFIC MORTALITY RATE

$$M_x^n = \frac{\text{\# of deaths between age } [x, x+n) \text{ and time } [0, T]}{\text{\# of person years lived between age } [x, x+n) \text{ and time } [0, T]}$$



$$m_x^n = \frac{d_x^n}{L_x^n} \rightarrow M_x^n$$

model

data

MODEL COHORT LIFE TABLE OF 10 PEOPLE

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Exact age x	Number left alive at age x l_x	Number dying between ages x and $x + n$ ${}_n d_x$	Probability of dying between ages x and $x + n$ ${}_n q_x$	Probability of surviving from age x to age $x + n$ ${}_n p_x$	Person-years lived between ages x and $x + n$ ${}_n L_x$	Person-years lived above age x $T_x = \sum_{a=x}^{\infty} {}_n L_a$	Expectation of life at age x $e_x^0 = T_x / l_x$	Death rate in the cohort between ages x and $x + n$ ${}_n m_x$	Average person-years lived in the interval by those dying in the interval ${}_n a_x$
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5	8	0	0	1	$8 \cdot 5 = 40$	$364.57 + 40 = 404.57$	$\frac{404.57}{8} = 50.571$	0	—
10	8	1	1/8	7/8	$7 \cdot 10 + 6.41 = 76.41$	$288.16 + 76.41 = 364.57$	$\frac{364.57}{8} = 45.571$	$\frac{1}{76.41}$	6.41
20	7	1	1/7	6/7	$6 \cdot 10 + 2.12 = 62.12$	$226.04 + 62.12 = 288.16$	$\frac{288.16}{7} = 41.166$	$\frac{1}{62.12}$	2.12

Preston, S., Heuveline, P., & Guillot, M. (2000). *Demography: measuring and modeling population processes*.

MEAN PERSON YEARS LIVED FROM MORTALITY

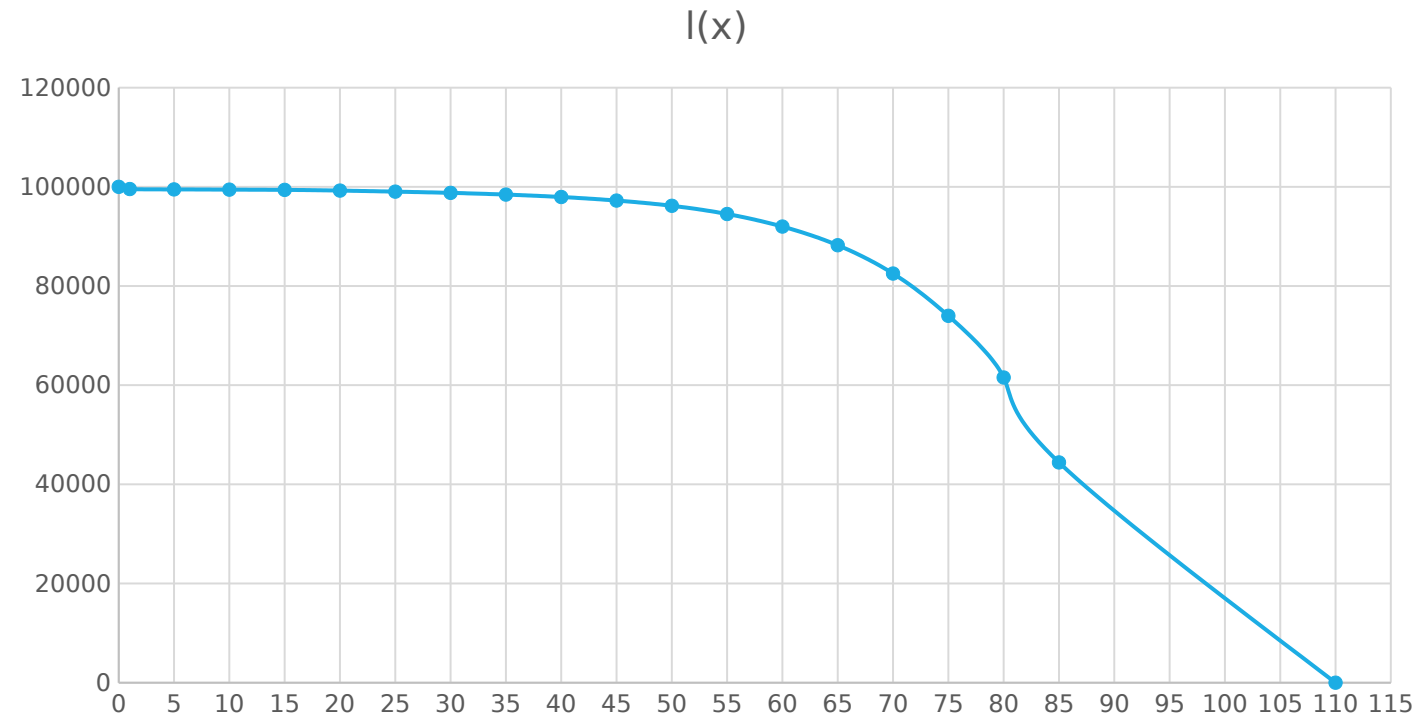
$$q(x, n) = 1 - e^{-n*m(x,n)}$$

$$p(x, n) = 1 - q(x, n)$$

$$l(x + n) = p(x, n) * l(x)$$

$$d(x, n) = q(x, n) * l(x)$$

$$L(x, n) = \left(\frac{l(x) + l(x+n)}{2} \right) * n$$



On excel: **$q(x,n)=1-\exp(-n*m(x,n))$**

DEATHS AND SURVIVAL

$$l(0) = 100,000$$

$l(x)$ = People surviving to age x

Deaths, between ages x and $x+n$
 $d(x, n) = l(x) - l(x + n)$

x	n	${}_nq_x$	${}_np_x$	l_x	${}_nd_x$	${}_nL_x$	T_x	e_x
0	1	0.008252	0.991748	100 000	825	99 258	7 756 261	77.563
1	4	0.001630	0.998370	99 175	162	396 311	7 657 003	77.207
5	5	0.000905	0.999095	99 013	89	494 842	7 260 692	73.331
10	5	0.000935	0.999065	98 924	93	494 388	6 765 850	68.394

$$q(x, n) = \frac{d(x, n)}{l(x)}$$

$$p(x, n) = 1 - q(x, n)$$

Probability of survival to age x

$$p(x) = \frac{l(x)}{l(0)}$$

DEATHS AND SURVIVAL

$$l(0) = 100,000$$

$l(x)$ = People surviving to age x

x	n	${}_nq_x$	${}_np_x$	l_x	${}_nd_x$	${}_nL_x$	T_x	e_x
0	1	0.008252	0.991748	100 000	825	99 258	7 756 261	77.563
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5	5	0.000905	0.999095	99 013	89	494 842	7 260 692	73.331
10	5	0.000935	0.999065	98 924	93	494 388	6 765 850	68.394

$$l(1) = ??$$

$$l(5) = ??$$

$$d(0,1) = ??$$

$$d(1,4) = ??$$

$$q(x,n) = \frac{d(x,n)}{l(x)}$$

$$p(x,n) = 1 - q(x,n)$$

$$p(1) = ??$$

$$p(5) = ??$$

$$p(10) = ??$$

PERSON YEARS- AREA UNDER THE CURVE

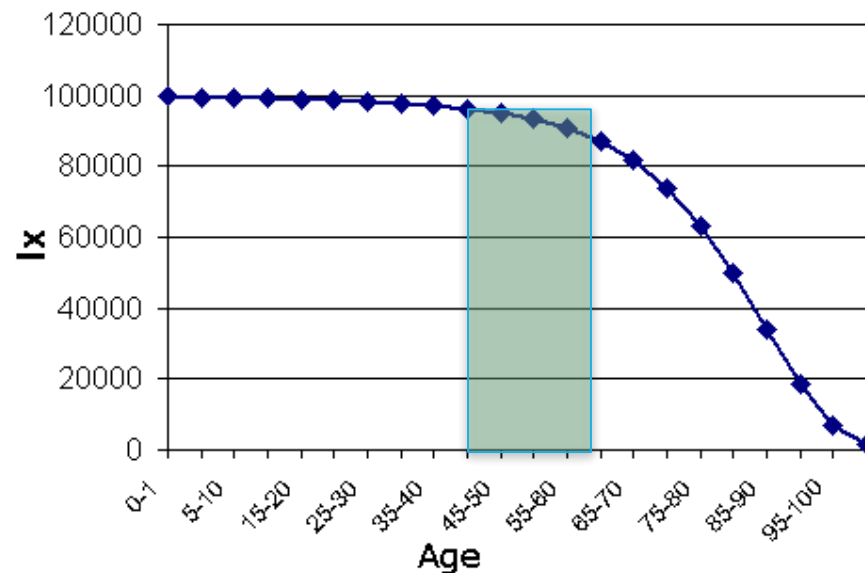
Person
Years

x	n	${}_nq_x$	${}_np_x$	l_x	${}_nd_x$	${}_nL_x$	T_x	e_x
0	1	0.008252	0.991748	100 000	825	99 258	7 756 261	77.563
1	4	0.001630	0.998370	99 175	162	396 311	7 657 003	77.207
5	5	0.000905	0.999095	99 013	89	494 842	7 260 692	73.331
10	5	0.000935	0.999065	98 924	93	494 388	6 765 850	68.394

$$L(x, n) = \left(\frac{l(x) + l(x + n)}{2} \right) * n$$

Older age
group

$$L_x^\infty = \frac{l_x}{m_x^\infty}$$



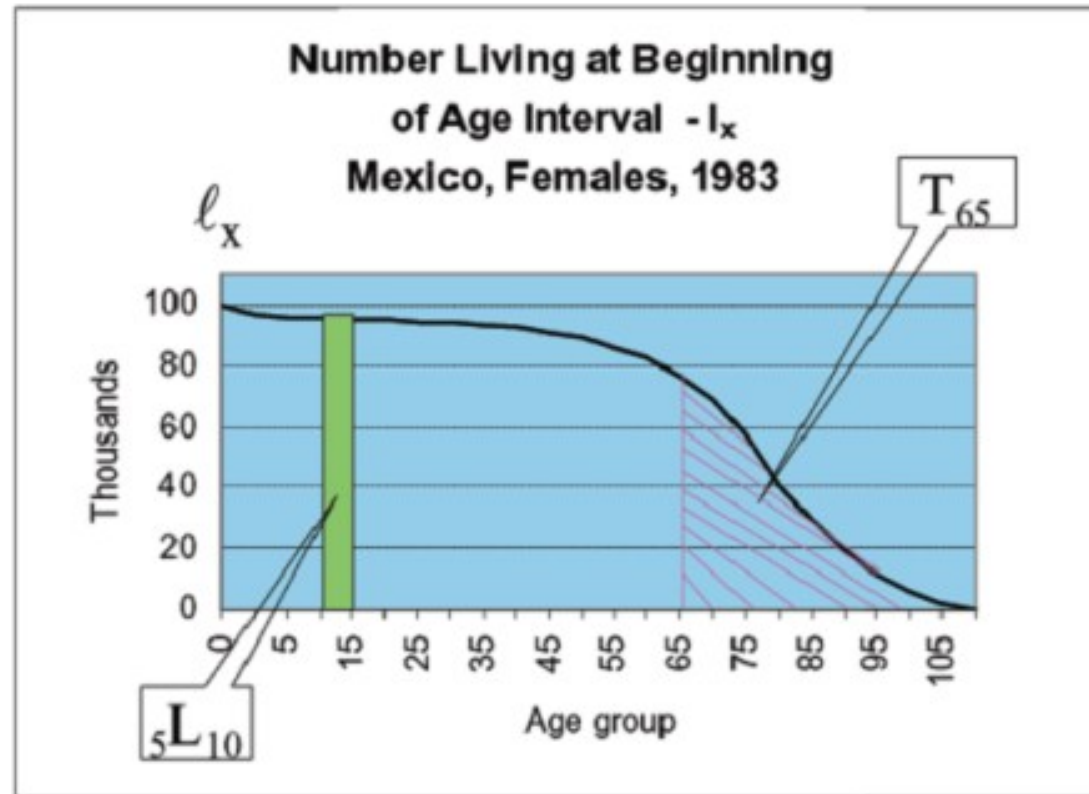
$$m(x, n) = -\frac{1}{l(x)} \frac{\Delta l(x)}{\Delta x}$$

=-slope/ $l(x)$

LIFE EXPECTANCY

$$T_x = \sum_{y=x}^{\infty} L_x^n = \int_x^{\infty} l(y) dy$$

$$e_x = \frac{T_x}{l_x}$$



Keyfitz and Fleiger ,
1990

INDIA – LIFE TABLE 2006-2010

Age-Interval x to x+n	${}_nq_x$	Total		
		l_x	${}_nL_x$	e_x
0-1	0.05227	100000	95781	66.1
1-5	0.01596	94773	375305	68.7
5-10	0.00598	93261	464909	65.8
10-15	0.00444	92703	462485	61.2
15-20	0.00683	92291	459981	56.4
20-25	0.00961	91661	456180	51.8
25-30	0.01050	90781	451572	47.3
30-35	0.01262	89828	446406	42.7
35-40	0.01647	88694	439910	38.3
40-45	0.01755	87233	432554	33.8
45-50	0.03318	85702	421873	29.4
50-55	0.04108	82859	406166	25.3
55-60	0.06097	79455	385987	21.3
60-65	0.10213	74611	355234	17.5
65-70	0.15611	66991	310042	14.2
70-75	0.23412	56534	250443	11.3
75-80	0.32714	43298	180901	9.0
80-85	0.42533	29133	113743	7.2
85+	...	16742	95820	5.7

EXERCISES

Life table calculation of countries
(Class example- INDIA)

STUDENT – LIFE TABLE EXERCISE

INTERPRETING LIFE TABLE

Box 3.2 Interpreting the Life Table

1. The basic columns:

x = Exact age x

l_x = Number of people left alive at age x

${}_n d_x$ = Number of people dying between ages x and $x + n$

${}_n q_x$ = Probability of dying between ages x and $x + n$

${}_n p_x$ = Probability of surviving from age x to age $x + n$

${}_n L_x$ = Person-years lived between ages x and $x + n$

T_x = Person-years lived above age x

e_x^o = Expectation of life at age x

${}_n m_x$ = Age-specific death rate between ages x and $x + n$

${}_n a_x$ = Average person-years lived between ages x and $x + n$ for persons dying in the interval

2. Additional information:

$(l_y)/l_x = {}_{y-x}p_x$ = Probability of surviving from age x to age y

$1 - (l_y)/l_x = {}_{y-x}q_x$ = Probability of dying between ages x and y

$l_x - l_y = {}_{y-x}d_x$ = Number of people dying between ages x and y

$T_x - T_y = {}_{y-x}L_x$ = Number of person-years lived between ages x and y

$({}_nd_x)/(l_0)$ = Probability that a newborn will die between ages x and $x + n$

$(l_x - l_y)/l_0$ = Probability that a newborn will experience his death between ages x and y

$(T_x - T_y)/l_0$ = Number of years that a newborn can expect to live between ages x and y

$$p(x) = \frac{l(x)}{l(0)}$$

COHORT LIFE TABLE OF MARRIAGE

Box 3.5 Application of Life Table Construction to Analysis of Marital Histories

Example: Bangladesh, cohort of 655 women aged 45–9 in 1993–4, who all married before age 30. State of interest: “never married”

Age x	n	l_x	nd_x	nq_x	na_x	nL_x	T_x	e_x^o
0	5	655	0					14.05
5	5	655	0					9.05
10	2	655	119					4.05
12	3	536	388					2.73
15	3	148	106					2.94
18	2	42	16					3.58
20	2	26	13					3.17
22	3	13	7					3.35
25	5	6	6					2.50
30								

Marital (Nuptial) Life

table

Interpretation of the columns:

n = length of interval

l_x = number of women who are never married at age x

${}_nd_x$ = first marriages between ages x and $x + n$

${}_nq_x$ = probability of having a first marriage between ages x and $x + n$

${}_na_x$ = average person-years lived in the never-married state between ages x and $x + n$ by women marrying in the interval (assumption: ${}_na_x = n/2$)

${}_nL_x$ = number of person-years lived in the never-married state between ages x and $x + n$

T_x = number of person-years lived in the never-married state above age x

e_x^o = expected number of years to be spent in the never-married state for a never-married person aged x . In this case, $e_0^o = 14.05$ can be interpreted as the mean age at marriage for those women.

Data source: Mitra, S. N., et al., 1994. *Bangladesh Demographic and Health Survey, 1993–1994*. Dhaka, Bangladesh, National Institute of Population Research and Training (NIPORT).

SINGLE DECREMENT LIFE TABLE IN OTHER PROCESSES

Table 3.7: *Examples of single decrement processes that can be studied by means of a life table^a*

<i>Process</i>	<i>State studied</i>	<i>State entered when...</i>	<i>State left when...</i>	<i>Vertical dimension of the table</i>
Mortality	Being alive	Born	Die	Duration of life (age)
Nuptiality (first marriage)	Being unmarried	Born	Marry	Duration of single life (age)
Migration from place of birth	Living in place of birth	Born	Move to another place	Duration of residence (age)
Entering the labor force	Having never worked	Born	First enter labor force	Duration of life (age)
Becoming a mother	Having no births	Born	Have first birth	Duration of life (age)
Subsequent childbearing	Not having an additional birth	Have a birth	Have an additional birth	Duration since having a birth
Marital survival	Being in intact marriage	Marry	Marriage ends	Duration of marriage
Unemployment spells	Being unemployed	Become unemployed	Leave state of unemployment	Duration of unemployment
Incarceration	Being in jail	Enter jail	Leave jail	Duration of incarceration

^a All of these processes can also be conceived as multiple decrement processes. Here we ignore other risks of leaving the state of interest.

GOMPERTZ-MAKHEM MORTALITY FUNCTION

$$\mu(x) = \alpha e^{\beta x} + \lambda$$

$$\beta \approx 0.085$$

